

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15E.7

¹H NMR SPECTROSCOPY

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E.7: ¹ H NMR Spectroscopy
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	4 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15E.7

Q1. 1. (a) Explain how D₂O (heavy water) is used in ¹H NMR to identify -OH and -NH protons (D₂O shake).

[3 Marks]

WCH14/01/Jan23/Q10(a)

Q2. (b) State what information is obtained from the integration trace (peak area ratio) in a ¹H NMR spectrum.

[1 Marks]

WCH14/01/Jan23/Q10(b)

SECTION B: WORKED MARK SCHEME — 15E.7

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Explain how D ₂ O (heavy water) is used in ¹ H NMR to identify -OH and -NH protons (D ₂ O shake). 1. Add D ₂ O to sample tube and shake. 2. Labile -OH / -NH protons undergo rapid deuterium exchange: -OH + D ₂ O → -OD + HOD. 3. Deuterium (² H) does not absorb in ¹ H NMR → the -OH / -NH peak disappears from the spectrum.	[3]
Q2	Q2: (b) State what information is obtained from the integration trace (peak area ratio) in a ¹ H NMR spectrum. Relative ratio of hydrogen atoms (protons) in each chemical environment.	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15E.7

FAQ #1: Why does the -OH peak disappear when D₂O is added?

Category: D₂O Exchange

COMMON EXAMINER TRAP / MISTAKE:

Stating D₂O reacts with the C=O group.

MODEL ANSWER / EXAMINER EXPECTATION:

Protons on -OH and -NH groups are acidic/labile and exchange rapidly with D in D₂O. Since D absorbs at a different frequency, the -OH peak vanishes.